

Introduction to Networks

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This presentation covers

- Introduction, Goals, means and challenges.
- General overview of Network Architecture & How it is mapped with the Syllabus.

Learning Outcome of this Module

- To understand the basic layers and its functions in Computer Networks.



What is Data Communication?

- Data - information presented in whatever form is agreed upon by the parties creating and using the data .
- Telecommunication - communication at a distance.
- Data communications → exchange of data between two devices via some form of transmission medium such as a wire cable.

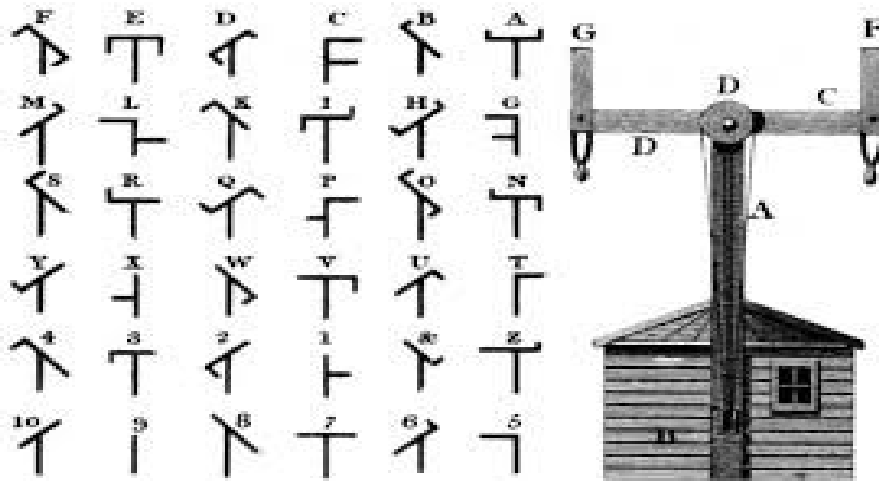
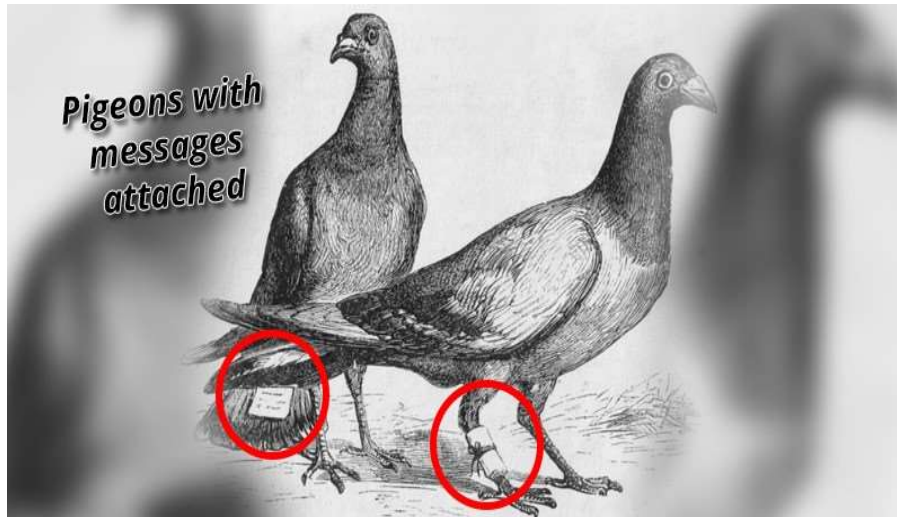


Fire / Smoke Signals



Talking Drum

Pre- Historic means of Data Communication



Hydraulic Semaphore

Optical Telegraph -1790's

Pre- Historic means of Data Communication



Heliograph – 1800's



Electric Telegraph – 1830's

Telephone -1870's



Network ?

- A network is a set of devices (often referred to as nodes) connected by communication links. A node can be a computer, printer, or any other device capable of sending and/or receiving data generated by other nodes on the network.

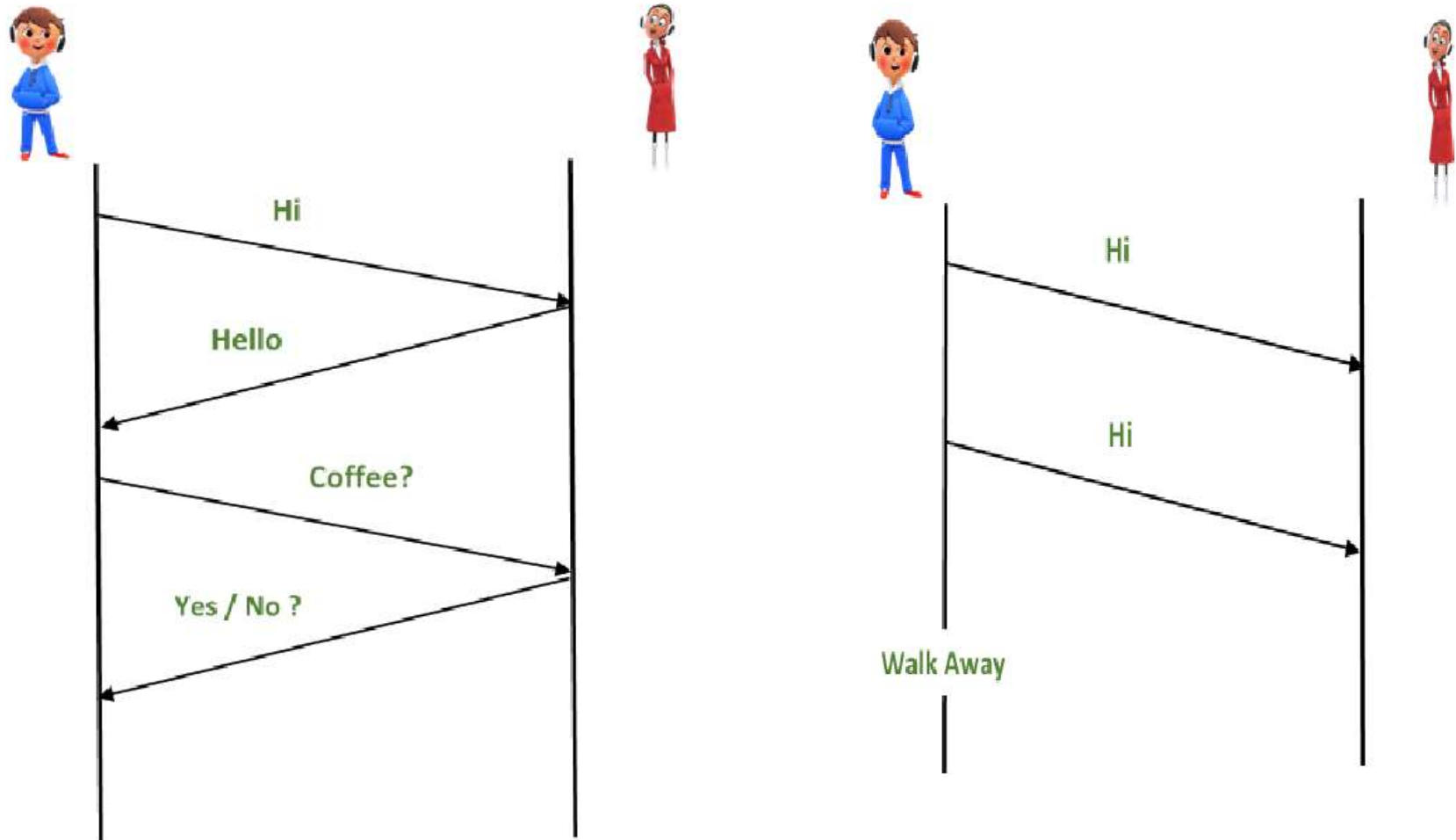
Goals

- Enable Efficient, Robust and Scalable Communication.
 - Efficient in terms of delay, cost etc.
 - Express mail
 - Regular mail
 - Robust towards failures or errors.
 - Scalable with more users and data.

Means

- Technology Development (Hardware).
 - Faster / Cheaper / Energy Efficient Routers, Links & Nodes.
- Protocols (Software): Implement many of the required functionality.
 - Provide reliability
 - Route packets
 - Share physical media etc...
 - Each protocol tries to achieve a specific goal.

Human Protocol

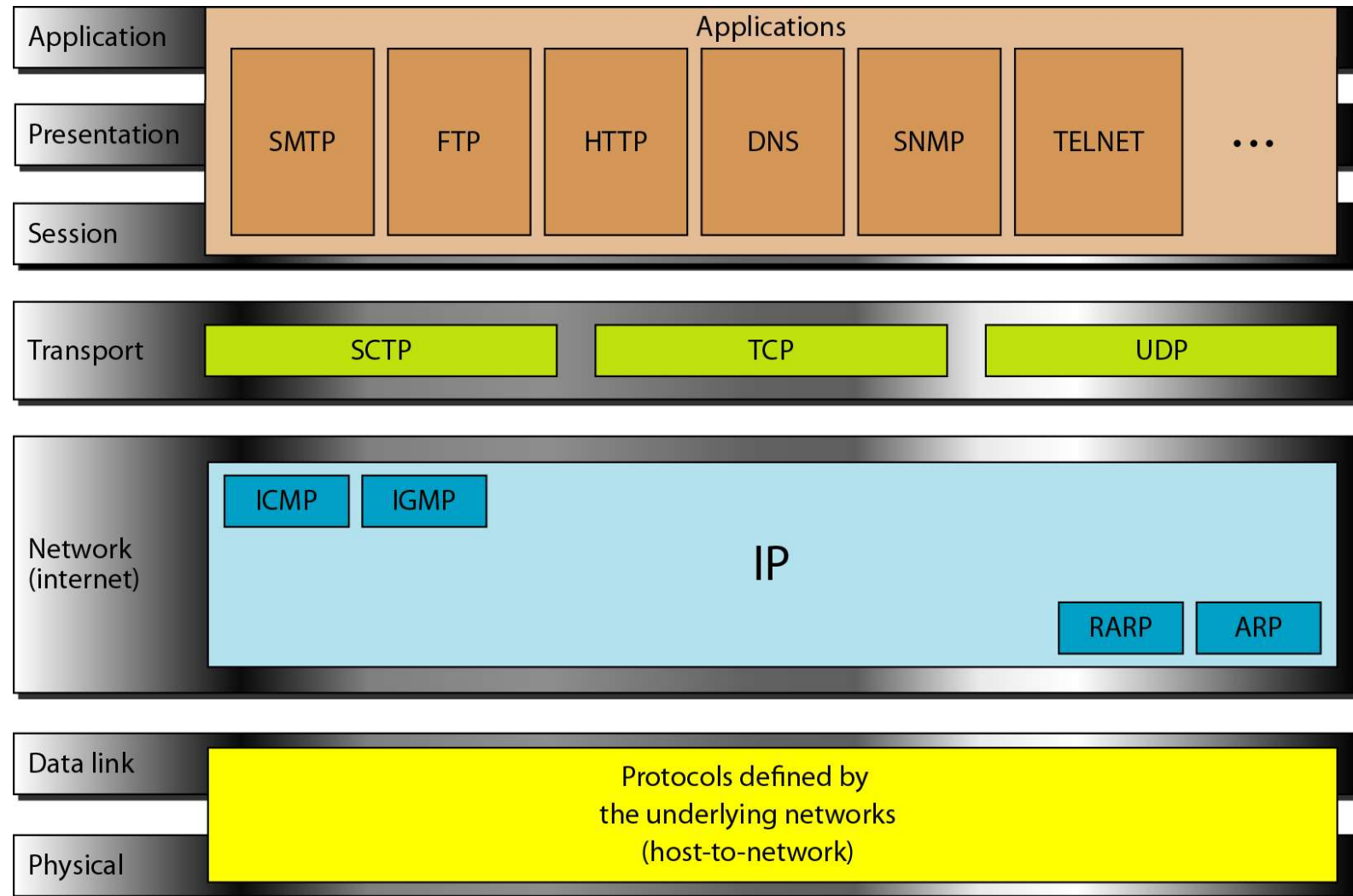


- What to sent? – Format of the message
- When to send & How to act ? – Rules
- Humans - Nodes

Challenges

- Tradeoff
 - Cost Vs Delay Vs Energy
 - Need to strike the right balance based on usage scenario.
- System Failure
 - Nodes can die, links corrupt packets, processing can duplicate, reorder packets.
- Backward Compatibility
 - Newer version of protocols should support older devices.

TCP/IP Reference model



Course Objectives

- To understand the protocol layering and physical layer communication.
- To understand the services of data link layer.
- To learn the functions of network layer and the various routing protocols.
- To familiarize the functions and protocols of the Transport layer and Application layer.

Thank You

